CS/SSW 555 Agile Methods

Cyber Guru Crew (Team 17)
Sprint 1
2/16/2024

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Project Background Information

Selected Project Option Number: Option 3

Main Objective of Project:
Develop a mobile application that uses AI to detect lies through voice stress, facial expressions, and response timing analysis. This application is aimed at professional and personal use, as well as it promising to enhance truth verification processes with user-friendly features.

Team Members and Roles:

Name	Role
Octavio Morales	Software Developer
Brandon Lopez	Machine Learning Analyst
Bhagawat Chapagain	Software Developer
Edmund Yuen	Machine Learning Analyst
Yunfei Tang	UX/UI Designer

List & Description of Each Sprint:

Sprint #	Objective
Sprint 1	Project onboarding and setup: ● Define database schema and set up the database ● Choose and set up testing framework ● Basic data collection for initial model training
Sprint 2	UI/UX development ● Design user interface for the app ● Implement user interactions and flow
Sprint 3	Machine Learning Model development: ● Select appropriate machine learning model ● Train the model with collected data ● Evaluate the performance of the model
Sprint 4	Machine Learning Model development: ● Fine-tune the machine learning model ● Integrate the UI components with the machine learning model

Project Links:

Type	URL
GitHub/GitLab/Etc	https://github.com/octavio-morales1/CS555Team17.git
Jira	https://stevens-team17.atlassian.net/jira/software/projects/KAN/boards/1#
Live Demo (if applicable)	N/A

Technology Stack Used:

Type	Technology
Data Preprocessing	MatLab
UI/UX	PyQt5 (Python GUI)
Machine Learning	Kaggle
Database	MongoDB

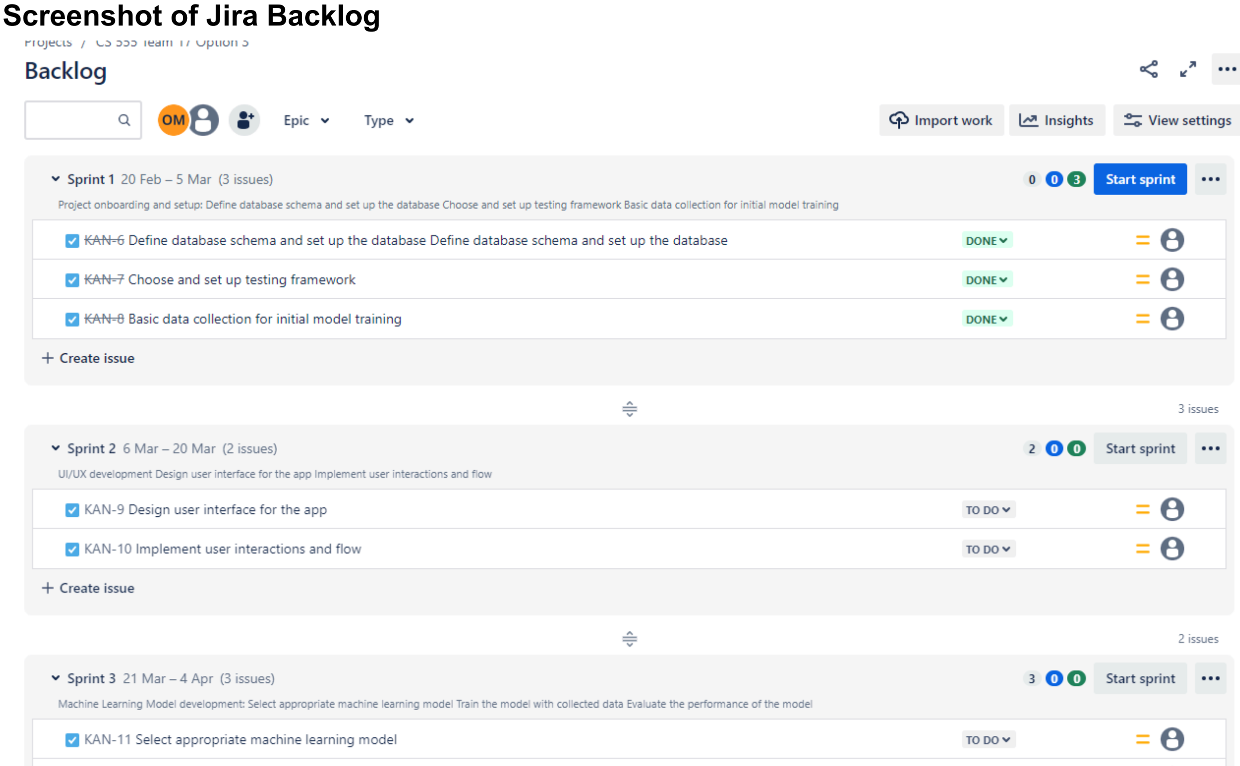
Sprint Progress Update: Current Status

- At our last meeting, we thought everything was being developed clearly, however, we noticed that we need to do a better job dividing the labor so that everyone is actually contributing towards the goal project.
- We want to continue using MongoDB because it allows us to use data from large datasets which is essential in developing a machine learning program
- We chose to, for now, test on a single variable **voice stress** which will be tested against some other threshold to indicate an emotion in the voice.
- We are in the process of collecting data through `.wav` search in open directory search engine ODCrawler. We’ll collect thousands of `.wav` files to be collected and categorized. Those of the human voice will be used to teach the model to detect stress in the voice.

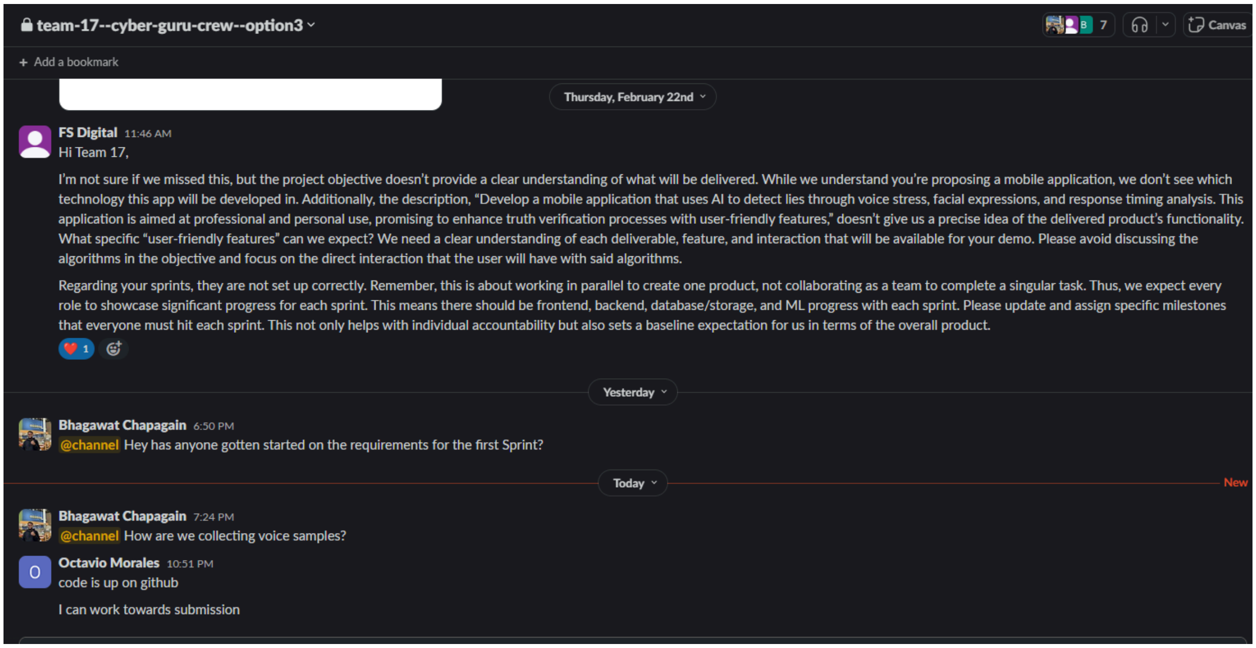
Summary of Work Completed

In the latest meeting, it was recognized that the division of labor needs improvement to ensure effective contribution towards the project goal. The project focuses on developing a machine learning program to detect stress in voices, utilizing MongoDB to manage large datasets. Currently, the team is concentrating on testing voice stress as a single variable against a threshold to identify emotions. Data collection is underway through .wav files obtained from the ODCrawler open directory search engine, which will be used to train the model in recognizing stress in human voices.

Screenshot of Jira Backlog



Slack Communication



Video of Project at Current State:

It does not need to be one edited video. If there are multiple videos, please link all of them below:

Public Video URL (Google Drive, Loom, Etc)
No video present as this is not the current stage for visual/audio results.

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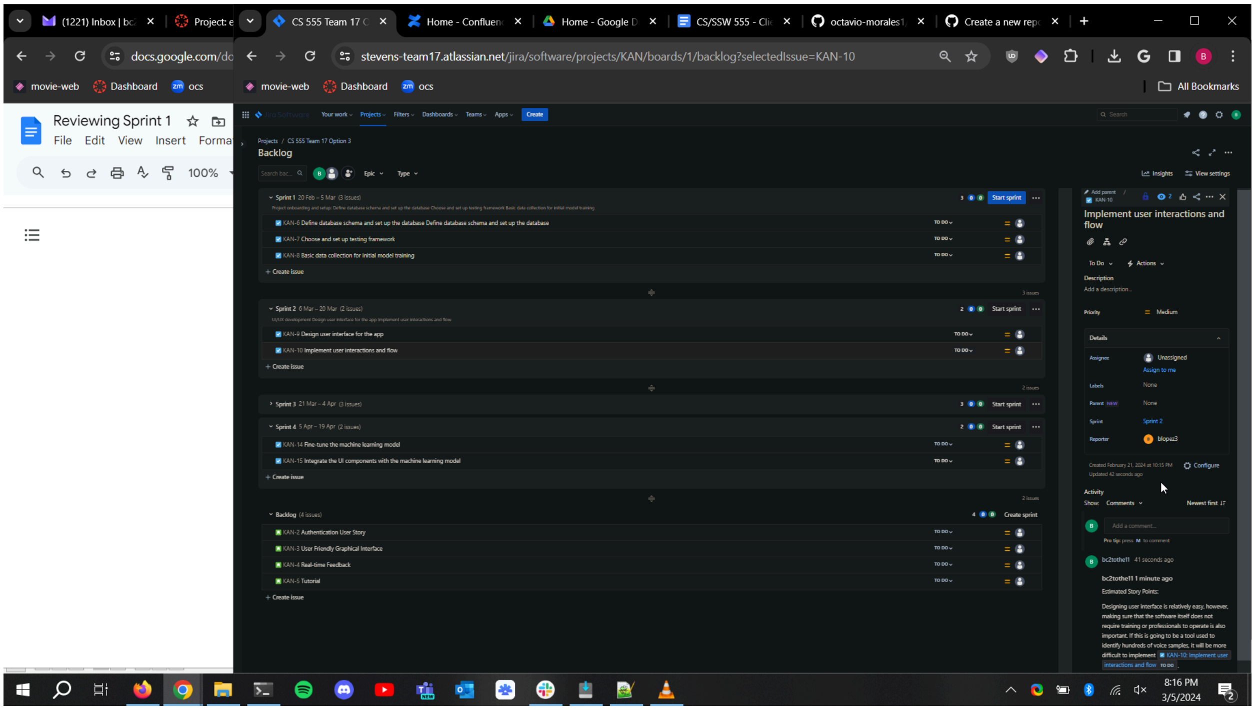
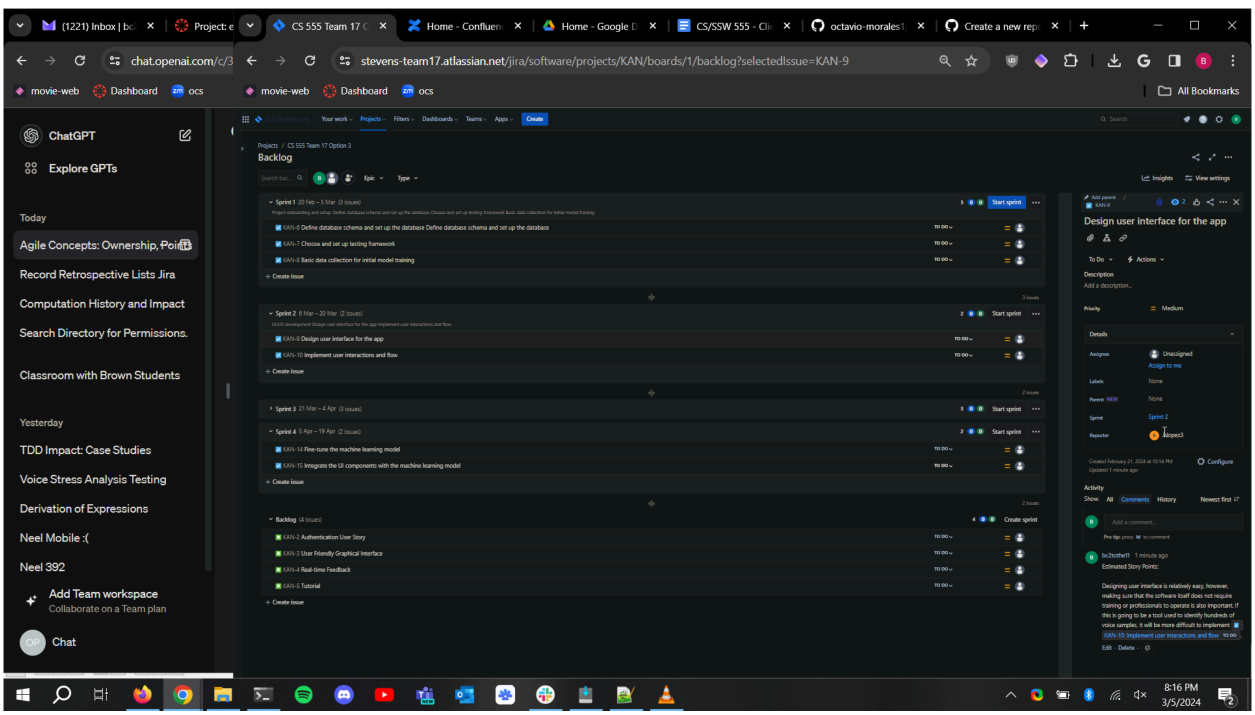
Sprint Progress Update: Current Blockers





Timeline Impacts:

Type	Impact

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<Database Connection>	<Adds 3 Days>

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Looking Ahead: Next Sprint

<Description or Bullet Points of Next Phase>

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